Venkata Lakshmi Parimala Pasupuleti

+1-703-261-5410 | venkatalakshmiparimalap@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Virginia, USA

PROFESSIONAL SUMMARY

Senior Software Engineer with 8+ years designing and owning production distributed systems, account-state platforms, and stateful data pipelines at scale. At Entrata, owned the core eligibility and financial settlement systems processing 180,000+ domain records for 100K+ monthly active users -- with data consistency guarantees, API design across microservices, service decomposition for scalability, and zero compliance violations across 7 years. Designed RabbitMQ-based async processing with idempotency guarantees for account state management, and led cloud migration of entitlements and document workflows to AWS S3 at 100% PHI compliance. Strong Python and Java backend engineering with distributed systems depth across microservices, caching (Redis), multi-tenant architecture, and cross-team platform delivery. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Distributed Systems + Data Consistency | Account State + Entitlements Platform | API Design + Service Decomposition | Stateful Data Pipelines at Scale | Financial Settlement + Credits | RabbitMQ Async + Idempotency Guarantees | Multi-Tenant Architecture + Data Integrity | Cross-Team Platform Delivery

WORK EXPERIENCE

Entrata (PropTech / Real Estate SaaS)

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Owned core eligibility and account state systems for 7 years** -- designed and maintained the annual data pipeline computing housing eligibility, entitlements, rent calculations, and financial settlement for 180,000+ domain records across 100K+ housing units. Data consistency and correctness were non-negotiable: a wrong eligibility decision affects thousands of residents. Zero compliance violations across all release cycles.
- **Designed RabbitMQ-based async processing system** for account state updates across thousands of tenant records -- retry logic and idempotency guarantees that eliminated double-processing failures; owned through production incidents and annual cycles. Service decomposition kept account state updates decoupled from blocking core application workflows.
- **Owned multi-tenant PostgreSQL schemas** for eligibility, financial settlement, and document workflows -- optimized 200+ SQL queries (30% gain) via execution plan analysis, indexing, partitioning, and stored procedure refactoring. Designed for API design consistency across microservices consuming these schemas.
- **Led cloud migration of entitlements and document workflows to AWS S3** -- inventoried all document types, mapped legacy records to target APIs, validated data consistency post-cutover, confirmed commerce platform workflows remained accurate. Libsodium-encrypted pipelines, 100% PHI compliance, zero security incidents.
- **Shipped 20+ full-stack platform modules** across 120+ Agile sprints; led microservices redesign with Docker and Kubernetes (OpenShift-compatible containers) -- 50% scalability improvement, 40% deployment time reduction; mentored 5+ engineers, 300+ peer code reviews.

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- Built agentic backend platform (LangGraph + FastAPI) with stateful workflow orchestration, API design across retrieval and validation services, and data consistency guarantees for compliance decisions. Integrated external APIs for regulatory data with validation guardrails.

George Mason University

Aug 2024 - Aug 2025

Research Associate · College of Engineering and Computing

- Built distributed data pipelines (Python, R) across 190+ countries -- multi-source ingestion, transformation, data consistency validation, and anomaly detection (Isolation Forest) across 3 research pipelines. R2=0.977. Designed Tableau and Power BI dashboards for stakeholder reporting.

Entrata India, Pune

Jul 2017 - Sep 2018

Associate Software Engineer

- Built account state and entitlements modules (Python, Java, PostgreSQL) for affordable housing; led cloud migration to AWS S3; redesigned data storage architecture migrating hundreds of thousands of records with full data consistency validation.

PROJECTS

Leasel: Distributed Account + Entitlements Platform for Compliance

Python · LangGraph · FastAPI · SQLite

Built enterprise platform managing account state, eligibility rules, and entitlement decisions across documents, settings, historical tickets, and regulatory data -- with data consistency guarantees (every decision cited or flagged). 8-agent orchestration runtime with conditional routing and HITL escalation for stateful account investigation. Results: 93.6% precision across 1,240 lease PDFs.
github.com/venkatalakshmiparimala/Lease_complaince_rag

AI on FHIR: API Platform for Medical Records

Flask · REST APIs · React · Docker · BERT

Built REST API platform translating natural-language queries into FHIR API calls across structured medical records -- HIPAA-compliant RBAC, data consistency across distributed records, ~60% reduction in manual retrieval time. github.com/venkatalakshmiparimala/AI_on_FHIR

AgentProbe: Platform Engineering with Evaluation Infrastructure

Python · FastAPI · CI/CD

Designed and built a 15-endpoint FastAPI backend platform with service decomposition across evaluation, routing, and reporting services -- CI-integrated quality gates, multi-provider API design, data consistency checks across eval runs. github.com/venkatalakshmiparimala/AgentProbe

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia — May 2026

GPA: 3.87

B.Tech Computer Science & Engineering JNTU Anantapur — 2017

SKILLS

Languages: Python, Java, Scala, JavaScript, TypeScript, SQL, PL/SQL, R, Bash

Backend & Distributed Systems: Distributed Systems, Account State Management, Entitlements Systems, Commerce Platform, API Design, Service Decomposition, Data Consistency, Financial Settlement, Credits and Discounts, RabbitMQ, Redis (caching), Microservices, Multi-Tenant Architecture, REST APIs, FastAPI, Spring Boot, gRPC

Cloud & Infra: AWS (S3, EC2, RDS, Glue), Azure (ADF, ADLS, Databricks, App Service), OpenShift, Docker, Kubernetes, Prometheus, Grafana, OpenTelemetry, CI/CD (GitHub Actions), Jira, Git

Databases: PostgreSQL, Oracle, SQL Server (T-SQL), MySQL, Snowflake, Teradata, Redis, NoSQL

AI & ML: LangChain, LangGraph, RAG, LLM Evaluation, HuggingFace, PyTorch, Scikit-learn, Pandas, NumPy